

Enhancing SINGER: Onboard Visual Drone Navigation through Iterative Imitation Learning and Reinforcement Fine-Tuning

Authors: Erwin Poussi, Maximilian Adang, Mac Schwager
Stanford Multi-Robot Systems Lab

Abstract

Autonomous drone navigation from onboard vision and natural language commands remains a challenging problem because the policy must simultaneously localize a semantically specified target, take collision-free actions, and output low-level rate commands, all from a single RGB stream without access to depth sensors or ground-truth positions. SINGER proposes a language-conditioned policy learned through behavioral cloning on expert RRT trajectories in a photorealistic 3D Gaussian Splatting environment, but yet still presents a gap to bridge since a comprehensive SINGER training achieves roughly 70% success with frequent collisions. We propose a pilot study of iterative DAgger enhancement of SINGER that bridges the distributional shift and multi-modal averaging of the BC-learned policy, combined with an improvement of the input representation by inferring visual goal-position features from CLIPSeg semantic segmentation, namely bearing and elevation angles to the target object.

We perform our study on a reduced pilot training dataset of approximately 150,000 samples compared to the 1.5 million used in SINGER’s canonical training. With our settings the SINGER baseline BC training achieves 36.7% success rate. When adding DAgger alone success rises to 52%, when combining BC with visually inferred goal-position features up to 80% with the exact same 150,000 training samples, and when pairing DAgger with visually inferred goal-position features up to 90% on average with sub-8% collision rate. These results demonstrate both the efficacy of DAgger but also the criticality of goal-position inference. Failure modes are mainly related to goal detection problems, which paves the way for improving reliable detection in future work and for potential reinforcement learning fine-tuning to overcome noisy spurious expert-gathered data.